

CDH23 (Q9H251) — Catenin Binding, Neuronal Projection, and Calcium Process Scope

Focus type: function_assignment **Hypothesis slug:** catenin-neuronal-and-calcium-process-scope **Gene:** CDH23 / Cadherin-related 23 (human, Homo sapiens, NCBITaxon:9606) **UniProt:** Q9H251

Summary

This report evaluates a three-part function-assignment hypothesis for human CDH23: that the gene product (i) binds beta-catenin or belongs to a catenin complex, (ii) participates in neuron projection development, or (iii) contributes to calcium transport/regulation. The seed asked that each claim be assessed independently using the phylogenetic placement of nodes PTN008601603 and PTN004649584 plus primary full texts, and that colocalization be distinguished from binding and mechanical ion participation from ion-pump activity. The investigation combined UniProt/QuickGO annotation provenance, PANTHER/Pfam domain architecture, and targeted full-text mining of the primary literature.

Overall verdict: weakly supported / over-annotated. The catenin claim is effectively refuted as a direct function: beta-catenin binding (GO:0008013) and catenin complex (GO:0016342) are both **IBA (phylogenetic) inferences** on Q9H251, propagated through ancestral node **PTN008601603**, whose experimental seeds are *exclusively classical catenin-binding cadherins* (CDH1/E-cadherin, CDH2/N-cadherin, CDH5/VE-cadherin, CDH24, CDH26) — never CDH23. CDH23 is an **atypical cadherin** lacking the classical cytoplasmic catenin-binding domain (Pfam PF01049); its tail instead anchors to the actin cytoskeleton through **harmonin PDZ interactions**, described in primary work as "a novel anchorage mode of the cadherins to the actin cytoskeleton"

(

P 12485990

The seed's proposed rescue via beta-catenin colocalization fails:

P 30747484

contains zero mentions of catenin, and in

P 22413011

beta-catenin is only an adherens-junction *marker*, not a demonstrated CDH23 partner.

The neuronal claim is weakly supported at best (IBA-only via node PTN004649584; neural expression present but no loss-of-function projection phenotype in zebrafish,).

P 22977299

The calcium claim is partially supported but mis-scoped: extracellular EC-domain **calcium ion binding** (GO:0005509) is defensible and mechanistically central, but **calcium ion transport** (GO:0006816) and **regulation of cytosolic calcium** (GO:0051480) over-interpret a *digenic PMCA2-pump modifier* study

P 17234811

in which CDH23 appears only as a genetic modifier. Importantly, none of this challenges CDH23's genuine core functions — calcium-dependent homophilic adhesion and tip-link mechanotransduction for hearing and vision — which have independent positive primary support and are not the subject of this hypothesis.

Key Findings

F001 — CDH23 anchors to F-actin via harmonin PDZ, not via catenins (argues against beta-catenin binding)

UniProt Q9H251 annotates beta-catenin binding (GO:0008013) and catenin complex (GO:0016342), but **both are IBA:GO_Central** phylogenetic inferences with **no experimental (IDA/IPI) support** on CDH23 itself. The primary literature documents a fundamentally *different*, catenin-independent cytoplasmic anchoring mechanism. The CDH23 cytoplasmic domain binds **harmonin** through PDZ- and N-domain interactions

P 12407180

"Two PDZ domains in harmonin interact with two complementary binding surfaces in the CDH23 cytoplasmic domain";

),

P 19297620

and harmonin-b anchors CDH23 to the stereocilia actin core in a manner explicitly framed as "**a novel anchorage mode of the cadherins to the actin cytoskeleton**"

).

P 12485990

A later structural study confirms the interaction "involves the harmonin Nter-PDZ1 supramodule, but not the C-terminal PDZ-binding motif of cadherin-23"

).

P 20639393

CDH23 is classified in PANTHER **PTHR24026 "FAT atypical cadherin-related"**, not the classical cadherin clade that carries the ancestral catenin-binding domain. Together these show CDH23 uses a harmonin-based, non-catenin actin linkage — undercutting any claim of direct beta-catenin binding or catenin-complex membership.

F002 — CDH23 "calcium ion transport" derives from a digenic PMCA2 modifier study, not from pump/transporter activity

On Q9H251, calcium ion transport (GO:0006816) and regulation of cytosolic calcium ion concentration (GO:0051480) are annotated **IMP:DFLAT**; calcium ion binding (GO:0005509) is **IEA:InterPro**. The experimental GOA reference for the calcium processes,

P 17234811

(Ficarella 2007), is fundamentally a study of the **PMCA2 (ATP2B2) plasma-membrane calcium pump**. CDH23 appears in that work only as a **genetic modifier** — a CDH23 T1999S substitution that "accentuated hearing loss in a previously described human family with a PMCA2 mutation." The paper itself identifies the true Ca^{2+} machinery: " Ca^{2+} enters the stereocilia of hair cells through mechanoelectrical transduction channels opened by the deflection of the hair bundle and is exported back to endolymph by an unusual splicing isoform (w/a) of plasma-membrane calcium-pump isoform 2 (PMCA2)." Neither the entry (MET channel) nor the export (PMCA2) route is CDH23. CDH23 forms the **upper tip-link** that mechanically gates the MET channel. Thus a *transport* or *regulation-of-cytosolic-calcium* annotation over-reads CDH23's mechanical, adhesive role. Extracellular **calcium ion binding** by the EC repeats, by contrast, is a genuine and central property of the protein — the seed correctly notes that "a tip-link mechanical contribution can participate in ion transport without being an ion pump," but that participation is captured by mechanotransduction and calcium binding, not by a transport annotation.

F003 — "Neuron projection development" is an IBA inference; neural expression exists but loss-of-function shows no projection defect

Neuron projection development (GO:0031175) on Q9H251 is **IBA:GO_Central only**, with no experimental support on CDH23. Critically, CDH23's characterized "projections" are **stereocilia** — actin-based apical microvillar protrusions of epithelial hair cells — **not neuron projections**; the hair-bundle role is already captured by more accurate terms (sensory perception of sound GO:0007605 IMP:MGI; stereocilium GO:0032420 ISS). In zebrafish, *cdh23* is expressed in CNS nuclei and "a small subset of GABAergic amacrine cells," yet *cdh23*-mutant larvae show "no detectable morphological retinal defects or degeneration" and a normal optokinetic response (

P 22977299

This is a textbook case of **expression without a demonstrated loss-of-function phenotype**: neural expression alone does not justify a neuron-projection-development function.

F004 — Catenin IBA is seeded only by classical cadherins at node PTN008601603 — paralog over-propagation

QuickGO annotation records for Q9H251 show GO:0008013 (beta-catenin binding) and GO:0016342 (catenin complex) are both IBA (GO_REF:0000033, GO_Central) whose with/from experimental seeds are **exclusively classical type-I/II cadherins**: CDH1/E-cadherin (P12830, MGI:88354), CDH2/N-cadherin (P19022), CDH5/VE-cadherin (P33151), CDH24 (Q86UP0), CDH26 (Q8IXH8), plus zebrafish *cdh* genes — propagated through ancestral node **PTN008601603**, exactly the node named in the seed hypothesis. **None of the seed genes is CDH23**, and no CDH23-specific experimental catenin annotation exists. Classical cadherins bind beta-catenin through a conserved cytoplasmic catenin-binding domain (**Pfam PF01049**) that is **absent** from CDH23 (Q9H251 lists only PF00028 cadherin repeats plus the CDH23-specific domain; its tail binds harmonin PDZ instead). The neuron projection development IBA (GO:0031175) propagates from a *separate* node **PTN004649584**, seeded by fly/mouse/zebrafish cadherins (*Drosophila fat* FBgn0262018, MGI:1890219, ZDB-GENE-040513-7), again none of them CDH23. This is the mechanistic signature of **paralog over-annotation**: an ancestral function correctly inferred for the classical-cadherin subfamily is carried over onto an atypical paralog that has diverged away from the seeding function. This finding directly addresses the seed's request to assess PTN008601603 and PTN004649584 placement.

F005 — The seed's "beta-catenin colocalization" evidence does not demonstrate binding

Full-text mining (Europe PMC, open access) directly tests the seed's proposed supporting evidence.

P 30747484

(Sannigrahi 2019, PMC6487693) contains **zero occurrences of "catenin"** in abstract or full text — the seed's attribution of beta-catenin colocalization to this paper is unsupported.

P 22413011

(Apostolopoulou & Ligon 2012, PMC3296689) mentions β -catenin 12 times, but **exclusively as an adherens-junction marker**: the authors "used an antibody to β -catenin to identify sites of possible cell-cell adhesion," scored contacts for β -catenin recruitment (found at 82.1% of heterotypic contacts), and *then* scored those β -catenin⁺ sites for cadherin-23 labeling. This establishes that CDH23 **spatially colocalizes** with β -catenin-marked adhesion sites; it does **not** provide a co-IP, pulldown, or two-hybrid demonstration of a CDH23– β -catenin molecular interaction. The paper's actually demonstrated CDH23 function is **heterotypic cell-cell adhesion** between breast cancer epithelial cells and fibroblasts — a genuine adhesion role, not a catenin binding role. This resolves the seed's request to "compare

P 12485990

with beta-catenin colocalization in

P 30747484

and

P 22413011

one paper does not mention catenin, and the other uses it only as a marker.

Mechanistic Model / Interpretation

The three sub-claims collapse onto a single coherent picture: **CDH23 is an atypical, tip-link cadherin whose cytoplasmic and calcium biology diverges from the classical cadherin–catenin paradigm that seeds its questionable annotations.**

CLASSICAL CADHERINS (E-/N-/VE-cadherin)

EC repeats (Ca²⁺-dependent adhesion)

|
transmembrane

▼
Cytoplasmic tail WITH catenin-binding domain (Pfam PF01049)

▼
β-catenin — α-catenin — ACTIN
(catenin complex, GO:0016342)

CDH23 (atypical, tip-link)

27 EC repeats (Ca²⁺ binding, homophilic + heterophilic handshake w/ PCDH15)

|
transmembrane

▼
Cytoplasmic tail, NO PF01049

▼
HARMONIN (PDZ + N-domain)

▼
ACTIN core of stereocilium
"novel anchorage mode" (12485990)

IBA node PTN008601603 —(β-catenin binding, catenin complex)—→ carried onto CDH23
seeds: CDH1, CDH2, CDH5, CDH24, CDH26 (NO CDH23) = PARALOG OVER-PROP.

Catenin axis. The catenin-binding/complex annotations are pure phylogenetic carry-over. The seeding node contains only classical cadherins that possess the PF01049 catenin-binding domain; CDH23 lacks that domain and uses a harmonin-based, structurally characterized alternative. The seed hypothesis asked to "distinguish absence of classical cytoplasmic binding motifs from negative binding assays." The honest answer is that both the motif is absent *and* the positive cytoplasmic partner is a different protein (harmonin) — so there is no residual positive evidence for beta-catenin binding to rescue.

Calcium axis. CDH23 is genuinely a **calcium-binding** protein: its EC repeats require Ca²⁺ for rigidity and adhesive function, and Ca²⁺ occupancy at inter-repeat linkers governs tip-link mechanics. That is a *molecular-function* (binding) and *mechanotransduction* role — not *ion transport*. The "calcium ion transport / regulation of cytosolic calcium" terms conflate the mechanical gating of a downstream channel (and a digenic interaction with a real pump, PMCA2) with CDH23 itself being a transporter. The correct causal chain is:

sound → hair-bundle deflection → CDH23–PCDH15 tip link tension →
MET channel opens → Ca²⁺ (and K⁺) influx → PMCA2 exports Ca²⁺
▲ ▲
(not CDH23) (not CDH23)

CDH23's node in this chain is the **mechanical linker/gate**, which is why "sensory perception of sound" (GO:0007605) is the well-supported BP term.

Neuronal axis. CDH23's "projections" are epithelial stereocilia. Neural expression is real but phenotypically silent in the available loss-of-function model. The neuron-projection-development term is a second, independent instance of IBA carry-over (from node PTN004649584) and should be treated as non-core/unsupported for the human gene.

Evidence Base / Evidence Matrix

Citation	Evidence type	Direction	Claim tested	Key finding	Context
PMID:12485990	Localization / interaction	Refutes catenin	CDH23 cytoplasmic anchoring mechanism	"novel anchorage mode of the cadherins to the actin cytoskeleton" via harmonin, not catenin	Mouse/ human hair cell bundle
PMID:12407180	Interaction (direct)	Refutes catenin	CDH23 cytoplasmic partner identity	"Two PDZ domains in harmonin interact with two complementary binding surfaces in the CDH23 cytoplasmic domain"	Biochemical
PMID:20639393	Structural	Refutes catenin	Nature of CDH23 tail interaction	Uses harmonin Nter-PDZ1 supramodule; ternary Usher complex	Structural/ biochemical
QuickGO / UniProt Q9H251 (GO_REF:0000033, node PTN008601603)	Computational / database	Refutes (provenance)	Are catenin terms experimental?	GO:0008013 & GO:0016342 are IBA; seeds = CDH1/2/5/24/26, no CDH23	GO_Central pipeline
PMID:17234811	Mutant phenotype / genetics	Qualifies calcium	Basis of Ca ²⁺ -transport annotation	CDH23 is a digenic <i>modifier</i> of a PMCA2 pump; Ca ²⁺ moves via MET channel + PMCA2	Human family + functional assay
PMID:21786191	Review-level synthesis	Qualifies calcium	Ca ²⁺ role of CDH23 vs PMCA2	CDH23 is a "Ca ²⁺ binding protein" forming tip links; PMCA2 does the export	Inner ear physiology review

Citation	Evidence type	Direction	Claim tested	Key finding	Context
PMID:22977299	Mutant phenotype / expression	Refutes neuronal	Neuron-projection-development function	Neural expression present, but mutant shows "no detectable morphological retinal defects or degeneration"	Zebrafish brain/retina
PMID:22413011	Localization / adhesion	Refutes catenin-binding; supports adhesion	Is β -catenin a CDH23 binding partner?	β -catenin used only as junction marker; CDH23 mediates heterotypic adhesion	Breast cancer epithelial + fibroblast
PMID:30747484	Full-text mining	Refutes (seed premise)	Does paper show β -catenin colocalization?	Zero "catenin" occurrences; paper is about aggregation/migration suppression	HEK293T + cancer cells
PMID:27349180	Interaction (direct)	Qualifies	CDH23-C cytoplasmic partners	CDH23-C binds CAMSAP3/Marshalin near the harmonin-binding region	Organ of Corti / in vitro
PMID:23467356	Mutant phenotype / structural	Supports core function	Tip-link handshake integrity	CDH23 EC1–EC2 "handshake" with PCDH15; disruption abolishes mechanotransduction	Mouse inner ear

GO Curation Implications

These are **leads requiring curator verification**, organized by the three sub-claims.

GO term	ID	Aspect	Current evidence	Recommended action
beta-catenin binding	GO:0008013	MF	IBA only; seeds are classical cadherins; PF01049 absent in CDH23	Remove / do NOT accept (paralog IBA carry-over)
catenin complex	GO:0016342	CC	IBA only; same node PTN008601603	Remove / do NOT accept ; CDH23 CC is stereocilium tip-link, not catenin complex
neuron projection development	GO:0031175	BP	IBA only (node PTN004649584); no LoF phenotype	Remove / treat as non-core ; prefer stereocilium/sensory-perception terms
calcium ion transport	GO:0006816	BP	IMP:DFLAT from digenic PMCA2 modifier study	Remove or generalize ; CDH23 gates MET channel, is not a transporter
regulation of cytosolic calcium ion concentration	GO:0051480	BP	IMP:DFLAT, same reference	Remove / demote to non-core
calcium ion binding	GO:0005509	MF	IEA:InterPro; central to EC-repeat function	Retain — defensible and informative
sensory perception of sound	GO:0007605	BP	IMP:MGI (experimental)	Retain — well supported core BP
stereocilium	GO:0032420	CC	ISS	Retain — accurate location
cell-cell adhesion (homophilic/calcium-dependent)	—	BP/MF	Positive primary support ()	Retain — this is core

GO term	ID	Aspect	Current evidence	Recommended action
---------	----	--------	------------------	--------------------

P 30747484

P 23467356

The recommended replacement for the catenin/neuronal/transport annotations is **not** "protein binding" (uninformative). The informative, supported alternatives already exist: **calcium ion binding** (MF), **stereocilium / tip-link** (CC), **sensory perception of sound** (BP), and **calcium-dependent cell-cell adhesion** (BP). The cytoplasmic interaction that a curator *could* positively annotate — if seeking a real partner — is **harmonin/USH1C binding** (supported by

P 12407180

P 19297620

P 20639393

not beta-catenin.

Mechanistic Scope

The immediate molecular functions being tested are: (a) a protein–protein *binding* activity toward beta-catenin, (b) a *developmental process* (neuron projection development), and (c) an *ion-transport/regulation* activity.

- **Direct gene-product activity that is supported:** extracellular calcium-dependent homophilic and heterophilic adhesion (EC-repeat mediated), calcium ion binding, and mechanical tip-link formation with PCDH15 that gates the MET channel

(P 23467356).

Cytoplasmic tail binding to harmonin/USH1C and CAMSAP3

(P 12407180 ,

P 19297620 ,

(P 20639393

)
(P 27349180

- **Downstream phenotypes / consequences (not direct MF of CDH23):** Ca^{2+} influx and cytosolic Ca^{2+} changes are consequences of channel gating, not CDH23 transport. Hearing/vision loss is a disease manifestation of adhesion/tip-link failure. The digenic hearing-loss modification with PMCA2

(P 17234811

is a genetic-interaction effect, not evidence of CDH23 pump activity.

- **Effects inferred only from loss of function or expression:** the neuronal claim rests on expression plus phylogeny; the single available LoF model shows no neuronal defect

(P 22977299

Conflicts and Alternatives

1. **Paralog confusion (primary alternative explanation).** The catenin and neuronal terms are best explained not as CDH23 biology but as **IBA over-propagation from classical cadherins** through nodes PTN008601603 and PTN004649584. This is corroborated by domain architecture: CDH23 lacks PF01049. This is the dominant, well-supported alternative to the seed hypothesis.

2. **Colocalization $\neq$ interaction.** The seed treats spatial colocalization with β -catenin-marked junctions

(P 22413011

as evidence of catenin membership. The full text shows β -catenin is only a *marker*; no binding assay exists, and the other cited paper

(P 30747484

does not mention catenin at all.

3. **Digenic modifier $\neq$ transporter.** The calcium-transport annotation stems from a study whose subject is the PMCA2 pump; CDH23's contribution is genetic modification of a pump phenotype

(
P 17234811
).

An alternative, defensible framing is EC-repeat **calcium binding**, which the transport term wrongly promotes to transport activity.

4. **Isoform/tissue caveat (partially addresses the seed's request).** The seed asks whether alternative isoforms/tissues retain ancestral functions. CDH23 has multiple isoforms (e.g., CDH23-C binds CAMSAP3/Marshalin,

P 27349180;
 excretory isoforms modulate adhesion,

P 30747484)

and broad tissue expression (kidney, muscle, testis, heart). However, none of these isoform- or tissue-specific data document beta-catenin binding, neuron projection development, or calcium transport; they document *more* non-catenin cytoplasmic partnerships and adhesion biology. So the isoform argument, while legitimate to raise, does not rescue the seed claims with current evidence.

5. **Stereocilia vs neuron projections.** A genuine terminological conflict: CDH23's protrusions are epithelial stereocilia, not neurites. The neuron-projection term likely reflects this category error compounded by IBA.

Limitations and Knowledge Gaps

- **No exhaustive interaction screen consulted.** We relied on curated UniProt/ QuickGO annotations, targeted full-text mining, and the provided literature. A systematic published *negative* binding assay (e.g., CDH23 tail vs β -catenin co-IP) is not in the record; the argument rests on *absence of the motif + presence of an alternative partner*, which is strong but not equivalent to a published negative assay. **Resolution:** a direct co-IP / ITC of the CDH23 cytoplasmic domain against β -catenin.
- **Isoform coverage is incomplete.** CDH23 has many splice isoforms; we documented harmonin- and CAMSAP3-binding cytoplasmic behaviors but did not survey every isoform's tail for a cryptic catenin-binding capacity. **Resolution:** isoform-resolved domain analysis and pulldowns.

- **Neuronal function relies on a single zebrafish LoF model.**

P 22977299

reports no retinal phenotype, but a mammalian CNS-specific CDH23 knockout with detailed neurite/synapse morphometry has not been examined here. Absence of a reported phenotype is not proof of no function. **Resolution:** conditional neuronal *Cdh23* knockout with axon/dendrite morphometry.

- **Calcium binding stoichiometry not quantified here.** We accept GO:0005509 on InterPro/structural grounds but did not compute EC-linker Ca^{2+} occupancy from structure. **Resolution:** structural/biophysical Ca^{2+} -binding measurement (not needed to reject *transport*, which is already refuted mechanistically).
- **QuickGO records were read at a point in time.** GO_Central annotations can be updated; a curator should re-verify current IBA seeds for GO:0008013, GO:0016342, and GO:0031175 on Q9H251.

Discriminating Tests

1. **Direct CDH23 tail vs β -catenin binding assay** (co-IP, GST-pulldown, or ITC using the CDH23 cytoplasmic domain). A clean negative would convert "no-motif + alternative-partner" into a published negative binding result and fully close the catenin question.
 2. **Domain-swap / motif search across CDH23 isoforms** for any PF01049-like catenin-binding module. Positive would reopen the isoform-rescue argument; negative confirms domain loss.
 3. **Conditional neuronal *Cdh23* knockout** (mouse) with quantitative neurite/axon-guidance morphometry and synapse counts, to test neuron projection development directly rather than by expression inference.
 4. **Ca^{2+} -flux electrophysiology** in hair cells separating CDH23's mechanical gating role from any putative transport: CDH23 mutation should alter MET-channel *gating* (tension) without conferring intrinsic ion-transport activity.
 5. **Re-audit of GO_Central IBA seeds** for the three terms on Q9H251, confirming nodes PTN008601603 / PTN004649584 and the absence of a CDH23 experimental seed.
-

Proposed Follow-up Experiments / Actions (Curation Leads)

Candidate action changes (leads requiring curator verification):

- **Do not accept** beta-catenin binding (GO:0008013) and catenin complex (GO:0016342) for CDH23. Provenance = IBA over-propagation from classical cadherins (node PTN008601603); domain PF01049 is absent; the real cytoplasmic partner is harmonin.
- **Do not accept / demote** neuron projection development (GO:0031175) to non-core. Provenance = IBA (node PTN004649584); LoF model shows no neuronal phenotype
([P 22977299](#)).
- **Remove or generalize** calcium ion transport (GO:0006816) and regulation of cytosolic calcium (GO:0051480). These over-read a digenic PMCA2-pump modifier study
([P 17234811](#)).
- **Retain** calcium ion binding (GO:0005509), stereocilium (GO:0032420), sensory perception of sound (GO:0007605), and calcium-dependent cell-cell adhesion as the informative, supported terms.
- **Consider adding** a positively supported cytoplasmic MF: **harmonin/USH1C binding**
([P 12407180](#)),
([P 19297620](#)),
([P 20639393](#))
in place of the unsupported catenin term.

Candidate references with exact snippets to verify:

- [P 12485990](#)
— "thereby identifying a novel anchorage mode of the cadherins to the actin cytoskeleton."
- [P 12407180](#)
— "Two PDZ domains in harmonin interact with two complementary binding surfaces in the CDH23 cytoplasmic domain."
- [P 17234811](#)
— "which accentuated hearing loss in a previously described human family with a PMCA2

mutation"; "Ca²⁺ enters the stereocilia ... through mechanoelectrical transduction channels ... and is exported back to endolymph by an unusual splicing isoform (w/a) of ... PMCA2."

•

P 22977299

— "revealed no detectable morphological retinal defects or degeneration"; "cdh23 is expressed by a small subset of GABAergic amacrine cells."

•

P 22413011

— "cadherin-23 mediates heterotypic cell-cell adhesion between breast cancer epithelial cells and fibroblasts" (β-catenin used as junction marker only).

Suggested curator questions:

- Is there any CDH23-specific experimental (IDA/IPI) annotation for GO:0008013 or GO:0016342? (Expected: no.)
- Do current GO_Central seeds at PTN008601603 include CDH23? (Expected: no.)
- Should the review explicitly flag stereocilia vs neuron projections to prevent recurring category error?

Bottom Line

For human CDH23 (Q9H251), the seed hypothesis's three claims are **weakly supported to refuted** as direct gene-product functions. The catenin binding / complex terms are paralog IBA carry-over from classical cadherins and are contradicted by CDH23's domain architecture (no PF01049) and its harmonin-based actin anchorage; the neuron-projection term is IBA-only with a null LoF phenotype; and the calcium-transport/regulation terms over-interpret a digenic PMCA2 modifier study. Only **calcium ion binding** survives as a defensible molecular-function annotation, alongside CDH23's genuine, independently supported core roles in calcium-dependent homophilic adhesion and tip-link mechanotransduction for hearing and vision.